UNIT V INTRUSION PREVENTION

Firewalls and Intrusion Prevention Systems: Need for Firewalls – Firewall Characteristics and Access Policy – Types of Firewalls – Firewall Basing – Firewall Location and Configurations – Intrusion Prevention Systems – Example Unified Threat Management Products.

5.1 Firewalls and Intrusion Prevention Systems

Firewall

A firewall is the first line of defense, installed at the network's perimeter to monitor and filter incoming and outgoing traffic. It uses security rules to evaluate data packets based on criteria like IP addresses and port numbers, and then allows, denies, or rejects them.

Intrusion prevention system

An IPS is a second line of defense that intercepts threats that get past the firewall. IPSs can monitor the entire network or a single host, and they can actively block threats.

A firewall is a network security device, either hardware or software-based, which monitors all incoming and outgoing traffic and based on a defined set of security rules it accepts, rejects or drops that specific traffic. **Accept :** allow the traffic **Reject :** block the traffic but reply with an "unreachable error" **Drop :** block the traffic with no reply A firewall establishes a barrier between secured internal networks and outside untrusted network, such as the Internet.

Before Firewalls, network security was performed by Access Control Lists (ACLs) residing on routers. ACLs are rules that determine whether network access should be granted or denied to specific IP address. But ACLs cannot determine the nature of the packet it is blocking. Also, ACL alone does not have the capacity to keep threats out of the network. Hence, the Firewall was introduced.

Connectivity to the Internet is no longer optional for organizations. However, accessing the Internet provides benefits to the organization; it also enables the outside world to interact with the internal network of the organization. This creates a threat to the organization. In order to secure the internal network from unauthorized traffic, we need a Firewall.

How does Firewall work?

Firewall match the network traffic against the rule set defined in its table. Once the rule is matched, associate action is applied to the network traffic. For example, Rules are defined as any employee from HR department cannot access the data from code server and at the same time another rule is defined like system administrator can access the data from both HR and technical department. Rules can be defined on the firewall based on the necessity and security policies of the organization. From the perspective of a server, network traffic can be either outgoing or incoming. Firewall maintains a distinct set of rules for both the cases. Mostly the outgoing traffic, originated from the server itself, allowed to pass. Still, setting a rule on outgoing traffic is always better in order to achieve more security and prevent unwanted communication. Incoming traffic is treated differently. Most traffic which reaches on the firewall is one of these three major Transport Layer protocols- TCP, UDP or ICMP. All these types have a source address and destination address. Also, TCP and UDP have port numbers. ICMP uses type code instead of port number which identifies purpose of that packet. **Default policy:** It is very difficult to explicitly cover every possible rule on the firewall. For this reason,

the firewall must always have a default policy. Default policy only consists of action (accept, reject or drop). Suppose no rule is defined about SSH connection to the server on the firewall. So, it will follow the default policy. If default policy on the firewall is set to accept, then any computer outside of your office can establish an SSH connection to the server. Therefore, setting default policy as drop (or reject) is always a good practice.

Generation of Firewall

Firewalls can be categorized based on their generation.

1. **First Generation- Packet Filtering Firewall:** Packet filtering firewall is used to control network access by monitoring outgoing and incoming packets and allowing them to pass or stop based on source and destination IP address, protocols, and ports. It analyses traffic at the transport protocol layer (but mainly uses first 3 layers). Packet firewalls treat each packet in isolation. They have no ability to tell whether a packet is part of an existing stream of traffic. Only It can allow or deny the packets based on unique packet headers. Packet filtering firewall maintains a filtering table that decides whether the packet will be forwarded or

   discarded. From the given filtering table, the packets will be filtered according to the following rules:

	Source IP	Dest. IP	Source Port	Dest. Port	Action
1	192.168.21.0	--	--	--	deny
2	--	--	--	23	deny
3	--	192.168.21.3	--	--	deny
4	--	192.168.21.0	--	>1023	Allow

Sample Packet Filter Firewall Rule

1. Incoming packets from network 192.168.21.0 are blocked.
2. Incoming packets destined for the internal TELNET server (port 23) are blocked.
3. Incoming packets destined for host 192.168.21.3 are blocked.
4. All well-known services to the network 192.168.21.0 are allowed.
5. **Second Generation- Stateful Inspection Firewall:** Stateful firewalls (performs Stateful Packet Inspection) are able to determine the connection state of packet, unlike Packet filtering firewall, which makes it more efficient. It keeps track of the state of networks connection travelling across it, such as TCP streams. So the filtering decisions would not only be based on defined rules, but also on packet's history in the state table.
6. **Third Generation- Application Layer Firewall :** Application layer firewall can inspect and filter the packets on any OSI layer, up to the application layer. It has the ability to block specific content, also recognize when certain application and protocols (like HTTP, FTP) are being misused. In other words, Application layer firewalls are hosts that run proxy servers. A proxy firewall prevents the direct connection between either side of the firewall, each packet has to pass through the proxy. It can allow or block the traffic based on predefined

rules. Note: Application layer firewalls can also be used as Network Address Translator(NAT).

7. **Next Generation Firewalls (NGFW):** Next Generation Firewalls are being deployed these days to stop modern security breaches like advance malware attacks and application-layer attacks. NGFW consists of Deep Packet Inspection, Application Inspection, SSL/SSH inspection and many functionalities to protect the network from these modern threats.

What is Magic Firewall?

"Magic Firewall" is a term used to describe a security feature provided by the web hosting and security company Cloudflare. It is a cloud-based firewall that provides protection against a wide range of security threats, including DDoS attacks, SQL injections, cross-site scripting (XSS), and other types of attacks that target web applications.

The Magic Firewall works by analyzing traffic to a website and using a set of predefined rules to identify and block malicious traffic. The rules are based on threat intelligence from a variety of sources, including the company's own threat intelligence network, and can be customized by website owners to meet their specific security needs.

The Magic Firewall is considered "magic" because it is designed to work seamlessly and invisibly to website visitors, without any noticeable impact on website performance. It is also easy to set up and manage, and can be accessed through Cloudflare's web-based control panel.

Overall, the Magic Firewall is a powerful security tool that provides website owners with an additional layer of protection against a variety of security threats.

5.2 Need of firewall

A firewall is a firmware or software that is an essential part of a computer network's security system. In simple terms, it acts as an intermediary or wall of separation between the insecure internet and secure internal network which may be a computer, company network, or home network. Separating the internet from your internal network traffic is the default function of most firewalls. However, you can change the default settings to allow selected trustworthy networks through the firewall thus creating controlled openings that don't compromise the network's safety but at the same time allow the passage of certain network traffic.

Importance of Using a Firewall :
The following points listed below are the most relevant in explaining the importance of firewalls is as follows.

Feature-1 :
Monitoring Network Traffic –
Firewall security starts with effective monitoring of network traffic based on pre-established rules and filters to keep the systems protected. Monitoring of network traffic involves the following security measures.

1. **Source or destination-based blocking of incoming network traffic –**
   This is the most common feature of most firewalls, whereby the firewalls block the incoming traffic by looking into the source of the traffic.

2. **Outgoing network traffic can be blocked based on the source or destination –**
   Many firewalls can also filter data between your internal network and the Internet. You might, for example, want to keep employees from visiting inappropriate websites.

3. **Block network traffic based on content –**
   More modern firewalls can screen network traffic for inappropriate content and block traffic depending on that. A firewall that is integrated with a virus scanner, for example, can prevent virus-infected files from entering your network. Other firewalls work in tandem with e-mail services to filter out unwanted messages.

4. **Report on network traffic and firewall activities –**
   When filtering network traffic to and from the Internet, it's also crucial to know what your firewall is doing, who tried to break into your network, and who tried to view prohibited information on the Internet. A reporting mechanism of some sort is included in almost all firewalls.

Feature-2 :
Stops Virus Attacks and spyware –
With cyber thieves creating hundreds of thousands of new threats every day, including spyware, viruses, and other attacks like email bombs, denial of service, and malicious macros, it's critical that you put protections in place to keep your systems safe. The number of entry points criminals can exploit to get access to your systems grows as your systems become more complicated and strong. Spyware and malware programs designed to penetrate your networks, manage your devices, and steal your data are one of the most common ways unwelcome persons obtain access. Firewalls are a crucial line of defense against malicious software.

Feature-3 :
Preventing Hacks –
Cyber threats are evolving at a fast pace and are widespread. Firewalls keep hackers out of your data, emails, systems, and other sensitive information. A firewall can either entirely block a hacker or push them to choose a more vulnerable target.

Feature-4 :
Promotes Privacy –
Having a firewall keeps the data safe and builds an environment of privacy that is trustworthy and a system without a firewall is accepting every connection into the network from anyone. Without a firewall, there would be no way to detect incoming threats. As a result, malicious users may be able to gain access to your devices and thereby compromising privacy. It's critical to take advantage of existing defenses to safeguard your network and the personal information stored on your computer against cybercrime.

5.3 Firewall Characteristics and Access Policy

Characteristics of Firewall

1. **Physical Barrier:** A firewall does not allow any external traffic to enter a system or a network without its allowance. A firewall creates a choke point for all the external data trying to enter the system or network and hence can easily block access if needed.

2. **Multi-Purpose:** A firewall has many functions other than security purposes. It configures domain names and Internet Protocol (IP) addresses. It also acts as a network address translator. It can act as a meter for internet usage.

3. **Flexible Security Policies:** Different local systems or networks need different security policies. A firewall can be modified according to the requirement of the user by changing its security policies.

4. **Security Platform:** It provides a platform from which any alert to the issue related to

security or fixing issues can be accessed. All the queries related to security can be kept under check from one place in a system or network.

5. **Access Handler:** Determines which traffic needs to flow first according to priority or can change for a particular network or system. specific action requests may be initiated and allowed to flow through the firewall.

Firewall Design Principles

1. Developing Security Policy

Security policy is a very essential part of firewall design. Security policy is designed according to the requirement of the company or client to know which kind of traffic is allowed to pass. Without a proper security policy, it is impossible to restrict or allow a specific user or worker in a company network or anywhere else. A properly developed security policy also knows what to do in case of
a security breach. Without it, there is an increase in risk as there will not be a proper implementation of security solutions.

2. Simple Solution Design

If the design of the solution is complex. then it will be difficult to implement it. If the solution is easy. then it will be easier to implement it. A simple design is easier to maintain. we can make upgrades in the simple design according to the new possible threats leaving it with an efficient but more simple structure. The problem that comes with complex designs is a configuration error that opens a path for external attacks.

3. Choosing the Right Device

Every network security device has its purpose and its way of implementation. if we use the wrong device for the wrong problem, the network becomes vulnerable. if the outdated device is used for a designing firewall, it exposes the network to risk and is almost useless. Firstly the designing part must be done then the product requirements must be found out, if the product is already available then it is tried to fit in a design that makes security weak.

4. Layered Defense

A network defense must be multiple-layered in the modern world because if the security is broken, the network will be exposed to external attacks. Multilayer security design can be set to deal with different levels of threat. It gives an edge to the security design and finally neutralizes the attack on the system.

5. Consider Internal Threats

While giving a lot of attention to safeguarding the network or device from external attacks. The security becomes weak in case of internal attacks and most of the attacks are done internally as it is easy to access and designed weakly. Different levels can be set in network security while designing internal security. Filtering can be added to keep track of the traffic moving from lower-level security to higher level.

5.4 Types of Firewalls

1.Packet-filtering

firewall

A Packet-filtering firewall filters all incoming and outgoing network packets. It tests them based on a set of rules that include IP address, IP protocol, port number, and other aspects of the packet. If the packet passes the test, the firewall allows it to proceed to its destination and rejects those that do not pass it.

Benefits of a Packet-filtering

- Quick and inexpensive
- Oldest and most fundamental firewall
- Protection against advanced threats is limited

2. Stateful Multi-Layer Inspection (SMLI)

Stateful Multi-Layer Inspection firewall employs packet inspection technology and TCP handshake verification to provide protection. These firewalls, also known as dynamic packet filtering, examine each network packet to determine whether it belongs to an existing TCP or another network session. The SMLI firewall creates a state table to store session information like source and destination IP address, port number, destination port number, etc.

Benefits of Stateful inspection

- Reduced traffic flow
- High-level protection
- Consumed significant system resources
- Provides extensive logging capabilities

3. Stateless firewall

Stateless firewalls monitor the network traffic and analyze each data packet's source, destination, and other details to determine whether a threat is present. These firewalls can recognize packet state and TCP connection stages, integrate encryption, and other essential updates.

Benefits of Stateless firewall

- Less complex
- Easy to implement
- Fast performance delivery
- Performs effectively in heavy traffic situations

4. Application-level gateway (Proxy firewall)

Application-level gateway, also called Proxy firewall, is used to protect data at the application level. It protects from potential internet hackers by not disclosing our computer's identity (IP address). Proxy firewalls analyze the context and content of data packets and compare them to a set of previously defined rules using stateful and deep packet inspection. They either permit or reject a package based on the outcome. Because this firewall checks the payload of received data packets, it is much slower than a packet-filtering firewall.

Benefits of Application-level gateways

- Safest firewall
- Deep packet inspection

- Significant slowdowns
- Safeguard resource identity and location

5. Circuit-level gateway

Circuit-level gateway validates established Transmission Control Protocol (TCP) connections. These firewalls typically operate at the OSI model's session level, verifying Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) connections and sessions. These firewalls are implemented as security software or as pre-installed firewalls. Like packet filtering firewalls, these firewalls do not examine the actual data packet but observe the information about the transaction.

Benefits of Circuit-level gateway

- Simple and inexpensive
- A single form of protection is insufficient
- Setup and management are simple

6. Next-Generation Firewall (NGFW)

The most common type of firewall available today is the Next-Generation Firewall (NGFW), which provides higher security levels than packet-filtering and stateful inspection firewalls. An NGFW is a deep-packet inspection firewall with additional features such as application awareness and control, integrated intrusion prevention, advanced visibility of their network, and cloud-delivered threat intelligence. This type of firewall is typically defined as a security device that combines the features and functionalities of multiple firewalls. NGFW monitors the entire data transaction, including packet headers, contents, and sources.

Benefits of Next-Generation Firewall

- Block malware
- Recognizing Advanced Persistent Threats (APTs)
- Less expensive
- Financially beneficial

7. Cloud firewall

A Cloud firewall, also known as FaaS (firewall-as-service), is a firewall that is designed using a cloud solution for network protection. Third-party vendors typically manage and operate cloud firewalls on the internet, and they are configured based on the requirements. Today, most businesses use cloud firewalls to protect their private networks or overall cloud infrastructure.

Benefits of Cloud firewall

- Unified security policy
- Flexible deployment
- Simplified deployment and maintenance
- Improved scalability
- Automatic updates

1. **Packet Filters –**
   It is a technique used to control network access by monitoring outgoing and incoming packets and allowing them to pass or halt based on the source and destination Internet Protocol (IP) addresses, protocols, and ports. This firewall is also known as a static firewall.

2. **Stateful Inspection Firewalls –**
   It is also a type of packet filtering which is used to control how data packets move through a firewall. It is also called dynamic packet filtering. These firewalls can inspect that if the packet belongs to a particular session or not. It only permits communication if and only if, the session is perfectly established between two endpoints else it will block the communication.

3. **Application Layer Firewalls –**
   These firewalls can examine application layer (of OSI model) information like an HTTP request. If finds some suspicious application that can be responsible for harming our network or that is not safe for our network then it gets blocked right away.

4. **Next-generation Firewalls –**
   These firewalls are called intelligent firewalls. These firewalls can perform all the tasks that are performed by the other types of firewalls that we learned previously but on top of that, it includes additional features like application awareness and control, integrated intrusion prevention, and cloud-delivered threat intelligence.

5. **Circuit-level gateways –**
   A circuit-level gateway is a firewall that provides User Datagram Protocol (UDP) and Transmission Control Protocol (TCP) connection security and works between an Open Systems Interconnection (OSI) network model's transport and application layers such as the session layer.

6. **Software Firewall –**
   The software firewall is a type of computer software that runs on our computers. It protects our system from any external attacks such as unauthorized access, malicious attacks, etc. by notifying us about the danger that can occur if we open a particular mail or if we try to open a website that is not secure.

7. **Hardware Firewall –**
   A hardware firewall is a physical appliance that is deployed to enforce a network boundary. All network links crossing this boundary pass-through this firewall, which enables it to perform an inspection of both inbound and outbound network traffic and enforce access controls and other security policies.

8. **Cloud Firewall –**
   These are software-based, cloud-deployed network devices. This cloud-based firewall protects a private network from any unwanted access. Unlike traditional firewalls, a cloud firewall filters data at the cloud level.

Comparison Table

Type	Layer	Key Feature	Best Use Case
Packet-Filtering Firewall	Network, Transport	Basic traffic filtering	Basic network security
Stateful Inspection Firewall	Transport	Tracks connection states	Context-aware security
Proxy Firewall	Application	Deep packet inspection	Application-level filtering
NGFW	Application (Multi-layer)	Integrated threat prevention	Comprehensive enterprise security
Circuit-Level Gateway	Session	Validates TCP/UDP sessions	Lightweight session security
Cloud Firewall	Application (Multi-layer)	Scalable, cloud-native security	Cloud environments
Hardware Firewall	Varies	High-performance dedicated appliance	Network perimeter security
Software Firewall	Varies	Flexible, host-based protection	Endpoint and host security
Virtual Firewall	Varies	Designed for virtualized environments	Virtual and cloud resource security

5.5 FIREWALL BASING

It is common to base a firewall on a stand-alone machine running a common oper- ating system, such as UNIX or Linux. Firewall functionality can also be imple- mented as a software module in a ro uter or LAN switch. In this section, we look at some additional firewall basing considerations.

Bastion Host

A bastion host is a system identified by the firewall administrator as a critical strong point in the network's security. Typically, the bastion host serves as a platform for an application- level or circuit-level gateway. Common characteristics of a bastion host are as follows:

- The bastion host hardware platform executes a secure version of its operating system, making it a hardened system.
- Only the services that the network administrator considers
- essential are installed on the bastion host. These could include proxy applications for DNS, FTP, HTT P, and SMTP.
- The bastion host may require additional authentication before a user is allowed access to the proxy services. In addition, each proxy
- service may require its own authentication before granting user access.
- Each proxy is configured to support only a subset of the standard application's command set.
- Each proxy is configured to allow access only to specific host systems. This means that the limited command/feature set may be applied only to a subset of

systems on the protected network.

- Each proxy maintains detailed audit information by logging all traffic, each connection, and the duration of each connection. The audit log is an essential
- tool for discovering and terminating intruder attacks.

- Each proxy module is a very small software package specifically designed for network security. Because of its relative simplicity, it is easier to check such
- modules for security flaws. For example, a typical UNIX mail application may contain over 20,000 lines of code, while a mail proxy may contain fewer than 1000.

- Each proxy is independent of other proxies on the bastion host. If there is a problem with the operation of any proxy, or if a future vulnerability is discov- ered, it can be
- uninstalled without affecting the operation of the other proxy applications. Also, if the user population requires support for a new service, the network administrator can easily install the required
- proxy on the bastion host.

- A proxy generally performs no disk access other than to read its initial config- uration file. Hence, the portions of the file system containing executable code
- can be made read only. This makes it difficult for an intruder to install Trojan horse sniffers or other d angerous files on the bastion host.

- Each proxy runs as a nonprivileged user in a private and secured directory on the bastion host.

Host-Based Firewalls

A host-based firewall is a software module used to secure an individual host. Such modules are available in many operating systems or can be provided as an add-on package. Like conventional stand-alone firewalls, host-resident firewalls filter and restrict the flow of packets. A common location for such firewalls is a server. There are several advantages to the use of a server-based or
workstation- based firewall:

• Filtering rules can be tailored to the host environment. Specific corporate security policies for servers can be implemented, with different
filters for servers used for different application.

• Protection is provided independent of topology. Thus both internal and exter- nal attacks must pass through the firewall.

• Used in conjunction with stand-alone firewalls, the host-based firewall pro- vides an additional layer of protection. A new type of server can be added to the network, with its own firewall, without the necessity of altering the net- work firewall configuration.

Personal Firewall

A personal firewall controls the traffic between a personal computer or workstation on one side and the Internet or enterprise network on the other side. Personal fire-
wall functionality can be used in the home environment and on corporate intranets. Typically, the pers onal firewall is a software module on the personal computer. In a home environment with multiple computers connected to
the Internet, firewall functionality can also be housed in a router that connects all of the home comput ers to a DSL, cable modem, or other Internet interface.

Personal firewalls are typically much less complex than either server-based firewalls or stand- alone firewalls. The primary role of the personal firewall is to deny
unauthorized remote access to the computer. The firewall can also monitor outgoing
activity in an attempt to detect and block worms and other malware.

An example of a personal firewall is the capability built in to the Mac OS X operating
system. When the user enables the personal firewall in Mac OS X, all inbound connections are denied except for those the user explicitly permits. Figure 22.2 shows this simple interface. The list of inbound services
that can be selectively reenabled, with their port numbers, includes the following:

- Personal file sharing (548, 427)
- Windows sharing (139)
- Personal Web sharing (80, 427)
- Remote login - SSH (22)
- FTP access (20-21, 1024-64535 from 20-21)
- Remote Apple events (3031)
- Printer sharing (631, 515)
- IChat Rendezvous (5297, 5298)
- ITunes Music Sharing (3869)
- CVS (2401)

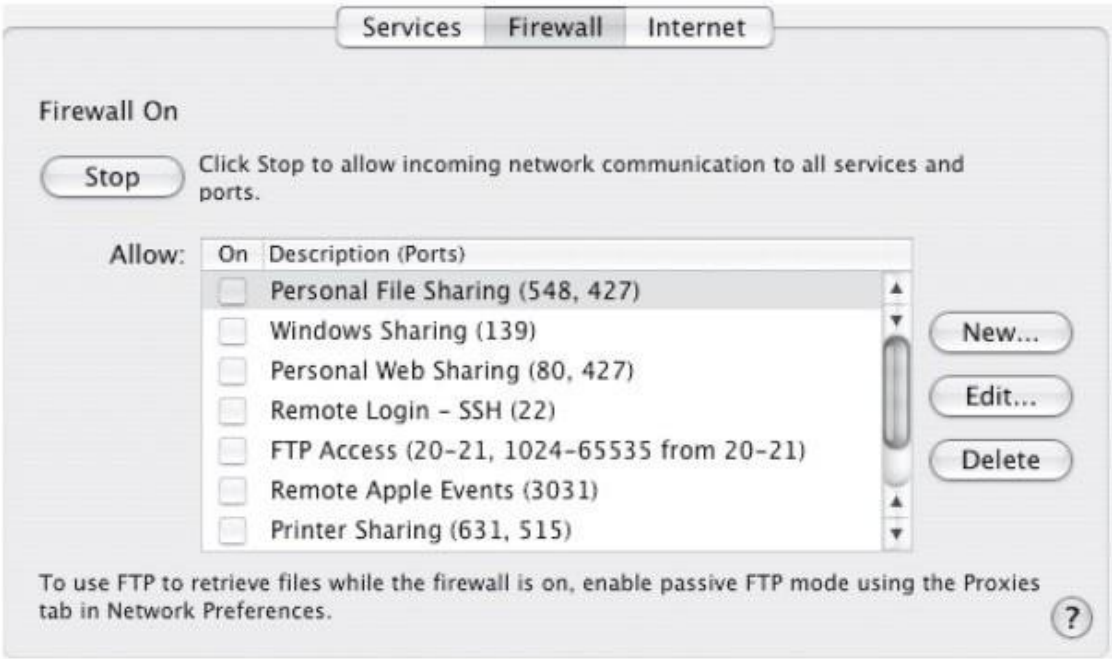

Figure 22.2 Example Personal Firewall Interface

- Gnutella/Limewire (6346)
- ICQ (4000)
- IRC (194)
- MSN Messenger (6891-6900)
- Network Time (123)
- Retrospect (497)
- SMB (without netbios-445)
- Timbuktu (407)
- VNC (5900-5902)
- WebSTAR Admin (1080, 1443)

When FTP access is enabled, ports 20 and 21 on the local machine are opened for FTP; if others connect to this computer from ports 20 or 21, the ports 1024 through 64535 are open.

For increased protection, advanced firewall features are available through easy-to-configure checkboxes. Stealth mode hides the Mac on the Internet by drop- ping unsolicited communication packets, making it appear as though no Mac is present. UDP packets can be blocked, restricting network traffic to TCP packets only for open ports. The fire wall also supports logging, an important tool for checking on unwanted activity.

5.6 FIREWALL LOCATION AND CONFIGURATIONS

As Figure 22.1a indicates, a firewall is positioned to provide a protective barrier between an external, potentially untrusted source of traffic and an internal network. With that general principle in mind, a security administrator must decide on the location and on the number of firewalls needed. In this section, we look at some common options.

DMZ Networks

Figure 22.3 suggests the most common distinction, that between an internal and an external firewall.
An external firewall is placed at the edge of a local or enterprise network, just inside the boundary router that connects to the Internet or some wide area network (WAN). One or more internal firewalls protect the bulk of the enter- prise network. Between these two types of firewalls are one or more networked devices in a region referred to as a DMZ (demilitarized zone) network. Systems that are externally accessible but need some protections are usually located on DMZ networks. Typically, the systems in the DMZ require or foster external con- nectivity, such as a corporate Web site, an e-mail server, or a DNS (domain name system) server.

The external firewall provides a measure of access control and protection for the DMZ systems consistent with their need for external connectivity. The external

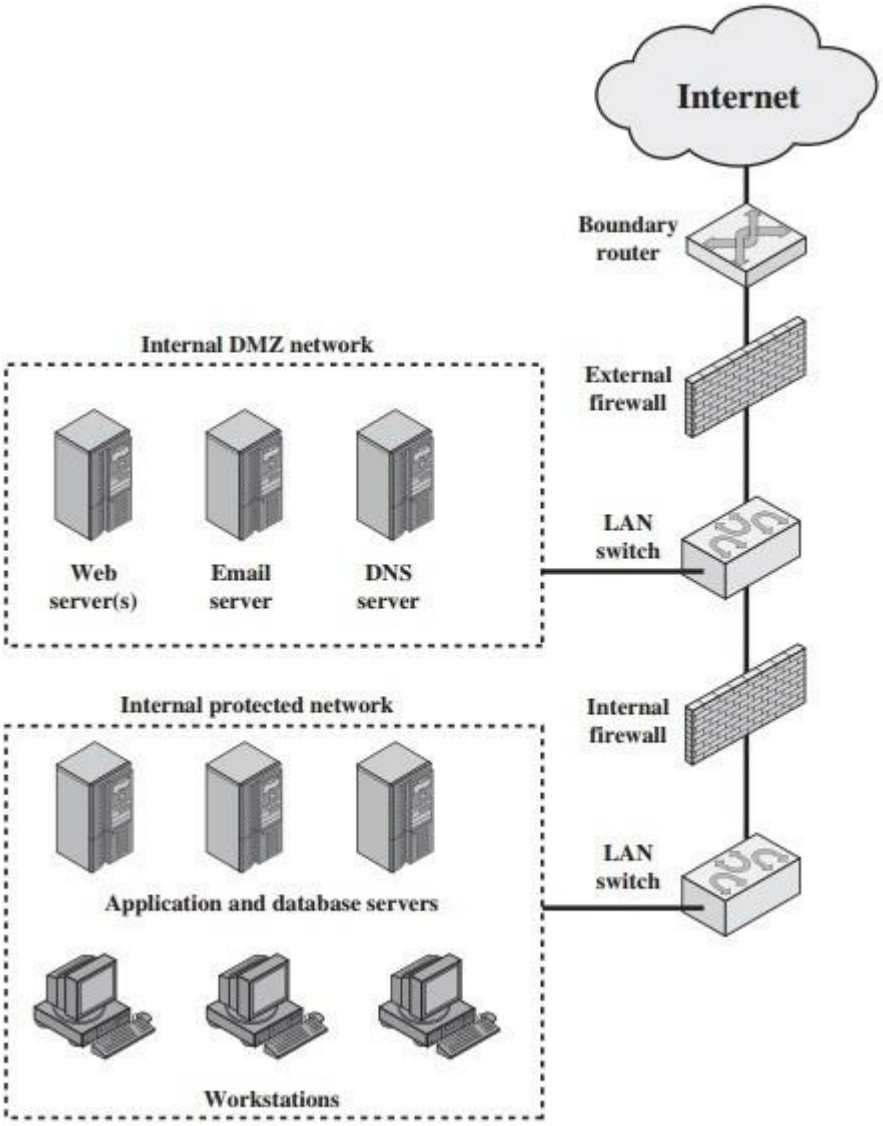

Figure 22.3 Example Firewall Configuration

firewall also provides a basic level of protection for the remainder of the enterprise network. In this type of configuration, internal firewalls serve three purposes:

1. The internal firewall adds more stringent filtering capability, compared to the external firewall, in order to protect enterprise servers and workstations from external attack.

2. The internal firewall provides two-way protection with respect to the DMZ. First, the internal firewall protects the remainder of the netw ork from attacks launched from DMZ systems. Such attacks might originate from worms, rootkits, bots, or other malware lodged in a DMZ system. Second, an internal firewall can protect the DMZ systems from attack from the inte rnal protected network.

3. Multiple internal firewalls can be used to protect portions of the internal network from each other. For example, firewalls can be configured so that internal servers are protected from internal workstations and vice versa. A common practice is to place the DMZ on a different network interface on the external firewall from that used to access the internal networks.

Virtual Private Networks

In today's distributed computing environment, the **virtual private network (VPN)** offers an attractive solution to network managers. In essence, a VPN consists of a set of computers that interconnect by means of a relatively unsecure network and that make use of encryption and special protocols to provide security. At each corporate site, workstations, servers, and databases are linked by one or more local area networks (LANs). The Internet or some other public network can be used to interconnect sites, providing a cost savings over the use of a private network and offloading the wide area network management task to the public network provider. That same public network provides an access path for telecommuters and other mobile employees to log on to corporate systems from remote sites.

But the manager faces a fundamental requirement: security. Use of a public network exposes corporate traffic to eavesdropping and provides an entry point for unauthorized users. To counter this problem, a VPN is needed. In essence, a VPN uses encryption and authentication in the lower protocol layers to provide a secure connection through an otherwise insecure network, typically the Internet. VPNs are generally cheaper than real private networks using private lines but rely on having the same encryption and authentication system at both ends. The encryption may be performed by firewall software or possibly by routers. The most common protocol mechanism used for this purpose is at the IP level and is known as IPsec.

An organization maintains LANs at dispersed locations. A logical means of implementing an IPsec is in a firewall, as shown in Figure 22.4, which essentially repeats Figure 19.1. If IPsec is implemented in a separate box behind (internal to) the firewall, then VPN traffic passing through the firewall in both directions is encrypted. In this case, the firewall is unable to perform its filtering function or other security functions, such as access control, logging, or scanning for viruses. IPsec could be implemented in the boundary router, outside the firewall. However, this device is likely to be less secure than the firewall and thus less desirable as an IPsec platform.

Distributed Firewalls

A distributed firewall configuration involves stand-alone firewall devices plus host- based firewalls working together under a central administrative control. Figure suggests a distributed firewall configuration. Administrators can configure host- resident firewalls on hundreds of servers and workstations as well as configure personal firewalls on local and remote user systems. Tools let the network adminis- trator set policies and monitor security across the entire network. These firewalls protect against internal attacks and provide protection tailored to specific machines and applications. Stand-alone firewalls provide global protection, including internal firewalls and an external firewall, as discussed previously.

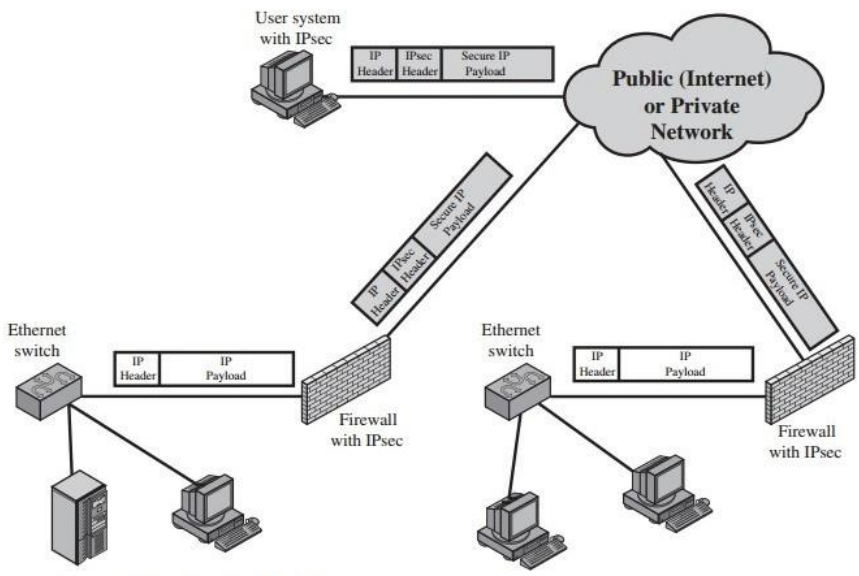

Figure 22.4 A VPN Security Scenario

With distributed firewalls, it may make sense to establish both an internal and an external DMZ. Web servers that need less protection because they have less critical information on them could be placed in an external DMZ, outside the exter- nal firewall. What protection is needed is provided by host-based firewalls on these servers.

An important aspect of a distributed firewall configuration is security monitoring. Such monitoring typically includes log aggregation and analysis, firewall statistics, and fine-grained remote monitoring of individual hosts if needed.

Summary of Firewall Locations and Topologies

We can now summarize the discussion from Sections 22.4 and 22.5 to define a spectrum of firewall locations and topologies. The following alternatives can be identified:

• **Host-resident firewall:** This category includes personal firewall software and firewall software on servers. Such firewalls can be used alone or as part of an in- depth firewall deployment.

• **Screening router:** A single router between internal and external networks with stateless or full packet filtering. This arrangement is typical for small office/home office (SOHO) applications.

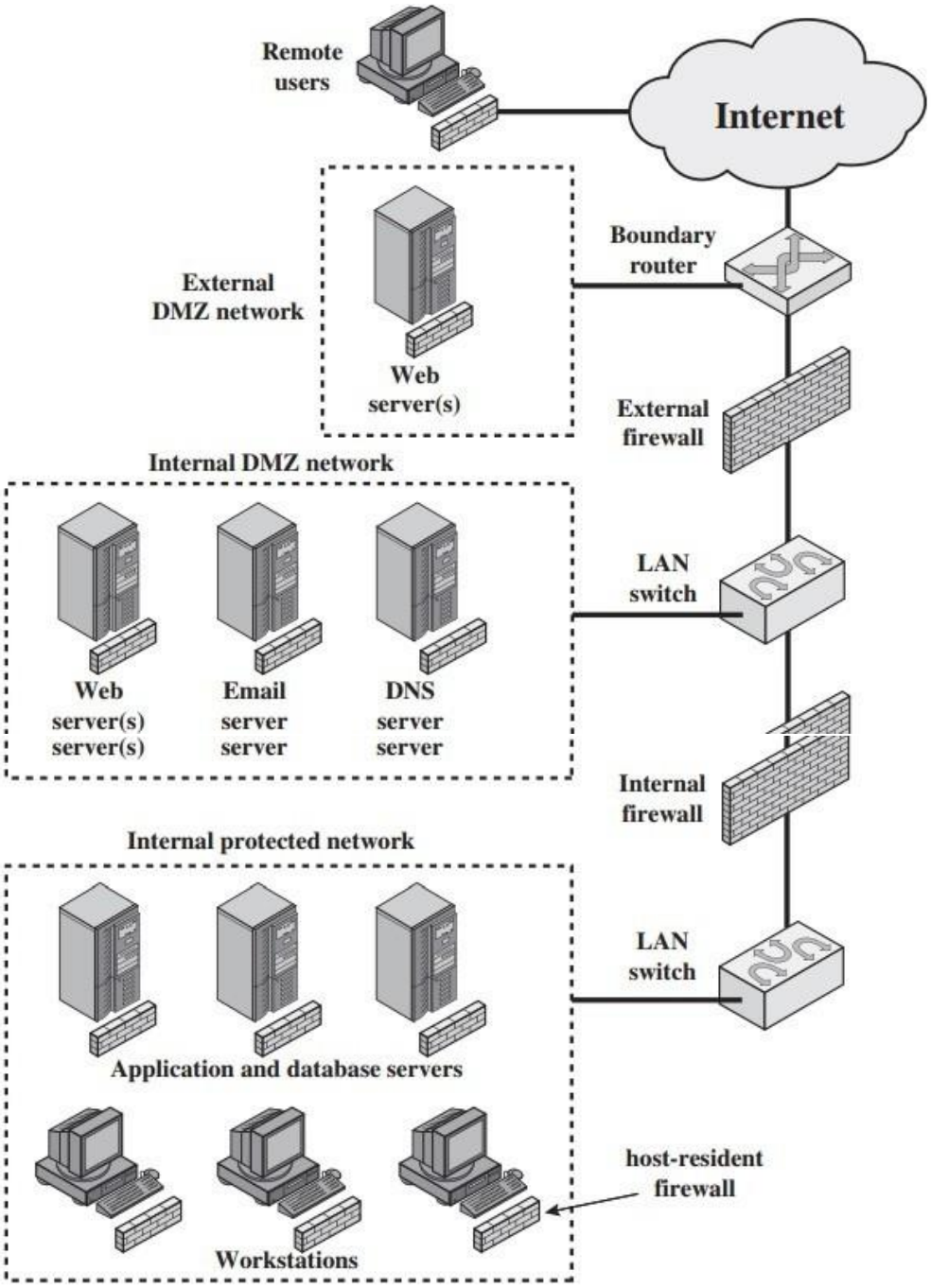

Figure 22.5 Example Distributed Firewall Configuration

Single bastion inline: A single firewall device between an internal and external router (e.g., Figure 22.1a). The firewall may implement stateful filters and/or application proxies. This is the typical firewall appliance configuration for small to medium-sized organizations.

• **Single bastion T:** Similar to single bastion inline but has a third network interface on bastion to a DMZ where externally visible servers are placed. Again, this is a common appliance configuration for medium to large organizations.

• **Double bastion inline:** Figure 22.3 illustrates this configuration, where the DMZ is sandwiched bet ween bastion firewalls. This configuration is common for large businesses and government organizations.

• **Double bastion T:** The DMZ is on a separate network interface on the bastion firewall. This configuration is also common for large businesses and govern- ment organizations and may be required. For example, this configuration is required for Australian government use (Australian Government Information Technology Security Ma nual - ACSI33).

• **Distributed firewall configuration:** Illustrated in Figure 22.5. This configura- tion is used by some large businesses and government organizations.

5.7 Intrusion Prevention System (IPS)

Intrusion Prevention System is also known as Intrusion Detection and Prevention System. It is a network security application that monitors network or system activities for malicious activity. Major functions of intrusion prevention systems are to identify malicious activity, collect information about this activity, report it and attempt to block or stop it.

Intrusion prevention systems are contemplated as augmentation of Intrusion Detection Systems (IDS) because both IPS and IDS operate network traffic and system activities for malicious activity.

IPS typically record information related to observed events, notify security administrators of important observed events and produce reports. Many IPS can also respond to a detected threat by attempting to prevent it from succeeding. They use various response techniques, which involve the IPS stopping the attack itself, changing the security environment or changing the attack's content.

How Does an IPS Work?

An IPS works by analyzing network traffic in real-time and comparing it against known attack patterns and signatures. When the system detects suspicious traffic, it blocks it from entering the network.

Types of IPS

There are two main types of IPS:

1. **Network-Based IPS:** A Network-Based IPS is installed at the network perimeter and monitors all traffic that enters and exits the network.
2. **Host-Based IPS:** A Host-Based IPS is installed on individual hosts and monitors the traffic that goes in and out of that host.

Why Do You Need an IPS?

An IPS is an essential tool for network security. Here are some reasons why:

- Protection Against Known and Unknown Threats: An IPS can block known threats and also detect and block unknown threats that haven't been seen before.
- Real-Time Protection: An IPS can detect and block malicious traffic in real-time, preventing attacks from doing any damage.
- Compliance Requirements: Many industries have regulations that require the use of an IPS to protect sensitive information and prevent data breaches.

- Cost-Effective: An IPS is a cost-effective way to protect your network compared to the cost of dealing with the aftermath of a security breach.
- Increased Network Visibility: An IPS provides increased network visibility, allowing you to see what's happening on your network and identify potential security risks.

Classification of Intrusion Prevention System (IPS):
Intrusion Prevention System (IPS) is classified into 4 types:

1. **Network-based intrusion prevention system (NIPS):**
   It monitors the entire network for suspicious traffic by analyzing protocol activity.

2. **Wireless intrusion prevention system (WIPS):**
   It monitors a wireless network for suspicious traffic by analyzing wireless networking protocols.

3. **Network behavior analysis (NBA):**
   It examines network traffic to identify threats that generate unusual traffic flows, such as distributed denial of service attacks, specific forms of malware and policy violations.

4. **Host-based intrusion prevention system (HIPS):**
   It is an inbuilt software package which operates a single host for doubtful activity by scanning events that occur within that host.

Comparison of Intrusion Prevention System (IPS) Technologies:
The Table below indicates various kinds of IPS Technologies:

IPS Technology Type	Types of Malicious Activity Detected	Scope per Sensor	Strengths
Network - Based	Network, transport, and application TCP/IP layer activity	Multiple network subnets and groups of hosts	Only IDPS which can analyze the widest range of application protocols;
Wireless	Wireless protocol activity; unauthorized wireless local area networks (WLAN) in use	Multiple WLANs and groups of wireless	Only IDPS able to predict wireless protocol activity

		clients	

NBA	Network, transport, and application TCP/IP layer activity that causes anomalous network flows	Multiple network subnets and groups of hosts	Typically more effective than the others at identifying reconnaissance scanning and DoS attacks, and at reconstructing major malware infections
Host-Based	Host application and operating system (OS) activity; network, transport, and application TCP/IP layer activity	Individual host	Can analyze activity that was transferred in end-to-end encrypted communications

Detection Method of Intrusion Prevention System (IPS):

1. **Signature-based detection:**
   Signature-based IDS operates packets in the network and compares with pre-built and preordained attack patterns known as signatures.

2. **Statistical anomaly-based detection:**
   Anomaly based IDS monitors network traffic and compares it against an established baseline. The baseline will identify what is normal for that network and what protocols are used.
   However, It may raise a false alarm if the baselines are not intelligently configured.

3. **Stateful protocol analysis detection:**
   This IDS method recognizes divergence of protocols stated by comparing observed events with pre-built profiles of generally accepted definitions of not harmful activity.

Comparison of IPS with IDS:

The main difference between Intrusion Prevention System (IPS) with Intrusion Detection Systems (IDS) are:

1. Intrusion prevention systems are placed in-line and are able to actively prevent or block intrusions that are detected.

2. IPS can take such actions as sending an alarm, dropping detected malicious packets, resetting a connection or blocking traffic from the offending IP address.

3. IPS also can correct cyclic redundancy check (CRC) errors, defragment packet streams, mitigate TCP sequencing issues and clean up unwanted transport and network layer options.

5.8 Unified Threat Management Definition

Unified threat management (UTM) refers to when multiple security features or services are combined into a single device within your network. Using UTM, your network's users are protected with several different features, including antivirus, content filtering, email and web filtering, anti-spam, and more.

UTM enables an organization to consolidate their IT security services into one device, potentially simplifying the protection of the network. As a result, your business can monitor all threats and security-related activity through a single pane of glass. In this way, you attain complete, simplified visibility into all elements of your security or wireless architecture.

Unified threat management (UTM) offers a product approaching total security in a box, ideal for small and midsize enterprises (SMEs). UTMs combine multiple network security functions in a single appliance. Typical solutions of UTM devices include intrusion prevention, antivirus, URL filtering, and VPN functionality.

Many products that were once labeled UTM are now marketed as firewalls, but they still serve a similar purpose. We've compiled our list of the six best UTM products for businesses and also provide a guide for buyers to narrow down their options and select the best product for their team.

- **SonicWall TZ Series Gen 7:** Best overall
- **WatchGuard Firebox M590/M690:** Best for value
- **Fortigate 900G:** Best for enterprises
- **Barracuda CloudGen Firewall F12A:** Best for public cloud management
- **Juniper Networks SRX2300:** Best for edge networks
- **Sophos XGS Desktop:** Best for SMBs looking to scale